

ProbOS — Month 1, Week 1

Distribution ABC

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Week 1 goal

Build the foundational `Distribution` abstract base class and concrete implementations (`Normal`, `LogNormal`, `Uniform`, `Beta`), in both Python and C++, with analytical `log_pdf` to avoid numerical underflow. This is the type system every later layer of ProbOS builds on.

Day-by-day plan

Day 1 – Distribution ABC design

- Define the abstract interface: `sample()`, `pdf()`, `log_pdf()`, `cdf()`, `ppf()`
- Decide: `log_pdf()` must be analytical, not `log(pdf(x))`, to avoid underflow on extreme values
- Set up `python/src/distributions.py` skeleton

Day 2 – Normal distribution

- Implement `Normal(mu, sigma)`: constructor validation (`sigma > 0`), `sample()` via NumPy's `default_rng`, analytical `pdf/log_pdf`
- Domain example baked into docstrings: battery SEI activation energy `Ea_SEI ~ Normal(1.35e5, 5e3)`
- First unit tests

Day 3 – LogNormal, Uniform, Beta

- `LogNormal(mu, sigma)` – parameters describe the underlying normal, not X itself
- `Uniform(low, high)` – validate `low < high`
- `Beta(alpha, beta)` – validate both shape parameters positive
- Constructor validation tests for all three

Day 4 – C++ Normal + build system

- Port Normal to C++20: `cpp/include/distributions/normal.hpp`, `cpp/src/distributions/normal.cpp`
- Set up `CMakeLists.txt`, Google Test integration
- First C++ unit tests

Day 5 – Test coverage + CI

- Fill out `python/tests/test_distributions.py`: sampling statistics, PDF/log-PDF agreement, boundary conditions
- Fill out `cpp/tests/test_normal.cpp`
- First working `.github/workflows/ci.yml`

Day 6-7 – Buffer / hardening

- mypy strict pass, ruff clean
- Documentation pass on all docstrings
- Confirm CI green end to end

What was actually accomplished (retrospective)

- Distribution ABC implemented exactly as planned: `sample()`, `pdf()`, `log_pdf()`, `cdf()`, `ppf()`
- Four concrete distributions: Normal, LogNormal, Uniform, Beta – all with fail-fast constructor validation and analytical `log_pdf`
- C++20 port of Normal with Google Test coverage
- 44 Python tests, 13 C++ tests – both suites passing
- `CMakeLists.txt` established with C++20 standard, Google Test integration guarded by `find_package(GTest QUIET)` so the build degrades gracefully without it
- First working CI pipeline (later extended across Weeks 2–6)

Numbers at Week 1 close

- Python tests: 44/44 passing
- C++ tests: 13/13 passing
- mypy strict: 0 errors
- ruff: 0 warnings

Carried into Week 2

The `Distribution ABC` is the foundation the `Model ABC` (Week 2) builds on directly – every uncertain physical parameter in `BatteryModel2Cell` is drawn from one of these four distribution types.